BCA 1st Semester - Practical Lab Assignments Guide

Assignment 2: Know Your Machine - Hardware + Windows OS Lab

Objective

After completing this assignment, you should be able to:

- Identify the hardware inside your Windows computer
- Check RAM, CPU, storage, and graphics information using Windows 11/10 Task Manager
- Use basic Windows Command Prompt commands
- Create and manage different user account types in Windows
- Understand how RAM and SSD affect Windows performance

Total Marks: 15 points

- Hardware information (5 marks)
- CMD commands + screenshots (5 marks)
- User account + performance explanation (5 marks)

Getting Started: Screenshot with Name + Roll No.

IMPORTANT: Every screenshot MUST clearly show your Name and Roll No.

Step 1: Open Notepad (Windows 11/10)

- Press **Windows Key** and type "notepad"
- Type your information:
    - Name: [Your Full Name]
    - Roll No.: [Your Roll Number]
- Keep this Notepad window visible in the corner when taking screenshots

Why? This proves you completed the work and prevents plagiarism accusations.

Part A: Know Your Hardware (5 marks)

What to Find

You'll use **Windows Task Manager** to find your computer's specifications.

Steps:

1. Press **Ctrl + Shift + Esc** to open Task Manager
2. Click the "Performance" tab
3. Select "CPU" to see processor information
4. Select "Memory" to see RAM information
5. Select "Disk" to see storage information
6. Select "GPU" to see graphics information

Information to Record

CPU (Processor):

- Model Name: _____
- Number of Cores: _____
- Logical Processors: _____

RAM (Memory):

- Total RAM: _____
- RAM Speed (MHz): _____

Storage (Disk):

- Disk Type (HDD/SSD): _____
- Total Disk Size: _____

GPU (Graphics):

- GPU Name: _____
- Dedicated Memory (if any): _____

Take a Screenshot:

- Capture the Task Manager Performance tab showing your hardware
- Make sure your Name + Roll No. (Notepad) is visible in the corner
- Save as: Screenshot_Hardware.png

Part B: Windows System Information (3 marks)

How to Access

Windows 11/10 stores detailed system information in the System Information tool.

Steps:

1. Press **Windows Key + R** (opens Run dialog)
2. Type: `msinfo32`
3. Press **Enter**
4. A "System Information" window will open

Information to Record

Find and write down:

- BIOS Version/Date: _____
- OS Name: _____
- OS Version: _____
- OS Build: _____
- System Manufacturer: _____
- System Model: _____

Take a Screenshot:

- Capture the System Information window
- Show your Name + Roll No. visible in corner
- Save as: Screenshot_SystemInfo.png

Part C: Windows Command Prompt Commands (5 marks)

What is Command Prompt?

Command Prompt (CMD) is a text-based interface for Windows. Instead of clicking, you type commands.

How to Open:

1. Press **Windows Key** and type "cmd" or "Command Prompt"
2. Click "Command Prompt" or "Windows Terminal"
3. A black/dark window appears

Commands to Run

Type each command and press **Enter**. Record what each shows:

1. whoami

- What it does: Shows your Windows username and domain
- Example output: COMPUTER\YourUsername
- Your output: _____

2. ipconfig

- What it does: Shows your network configuration (IP address, gateway, DNS)
- Look for: IPv4 Address
- Your IP Address: _____
- Your Default Gateway: _____

3. dir

- What it does: Lists all files and folders in current directory
- Record 3 folders you see: _____, _____, _____

4. mkdir BCA_Test

- What it does: Creates a new folder named "BCA_Test"
- Verify by typing: dir
- Did you see "BCA_Test" folder created? Yes / No

5. tasklist

- What it does: Lists all running programs
- Record 5 running processes: _____, _____, _____, _____, _____

6. systeminfo

- What it does: Shows detailed Windows system information
- Record your Processor: _____
- Record your Total Physical Memory: _____

Take Screenshots:

- Capture each command and its output
- Make sure Name + Roll No. visible in each screenshot
- Save as: Screenshot_CMD_1.png, Screenshot_CMD_2.png, etc.

Part D: Create a Standard User Account (4 marks)

Why Different User Accounts?

Windows has three main account types:

- **Administrator:** Full control, can install software, change system settings
- **Standard User:** Limited access, cannot install most software, cannot change system settings
- **Guest:** Very limited, temporary access only

How to Create a Standard User in Windows 11/10

Steps:

1. Go to **Settings** (Press Windows Key + I)
2. Click **Accounts**
3. Click **Other users** (or "Family & other users")
4. Click **Add account**

5. Click **I don't have this person's sign-in information**
6. Click **Add a user without a Microsoft account**
7. Enter:
    - Username: TestUser
    - Password: [Create a simple password]
    - Confirm password
8. Click **Next**
9. Make sure **"Standard user"** is selected (NOT Administrator)
10. Click **Finish**

Test the Standard User Account

Log in with the new Standard User:

1. Sign out of your current account
2. Click the TestUser account
3. Enter the password
4. You're now logged in as a Standard User

Try to install software:

1. Try downloading and installing any software (e.g., VLC, 7-Zip)
2. You'll get a permission dialog asking for Administrator password
3. You cannot proceed without Administrator permission

Questions to Answer

Q1: Can a Standard User install every type of software? Answer: _____

Q2: What happens when administrator permission is required? Answer: _____

Q3: What is UAC? Explain in 2-3 sentences. Answer: _____

UAC Hint: UAC = User Account Control. It's a Windows security feature that asks for permission before allowing important changes to your computer.

Take Screenshots:

- Screenshot showing the new Standard User account created
- Screenshot showing the permission dialog when trying to install software
- Save as: Screenshot_StandardUser.png, Screenshot_Permission.png

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Assignment 1 Submission Checklist

- ☐Part A: Hardware information completed
- ☐Part B: Windows System Information completed
- ☐Part C: All 6 CMD commands executed and results recorded
- ☐Part D: Standard User created and tested
- ☐Screenshots taken with Name + Roll No. visible
- ☐No plagiarism - answers are in your own words
- ☐PDF created with all answers and screenshots

Submission

File name format: RollNo_Name_Assignment1.pdf

Example: BCA001_Rahul_Kumar_Assignment1.pdf

Submit as: PDF with 3-4 pages including all screenshots and answers